Fiscal Year:	FY 2006	**Task Last Updated:**	FY 05/09/2006
PI Name:	Sundaresan, Alamelu Ph.D.		
Project Title:	Targeting specific bone cell signaling pathways and immunosuppression in microgravity		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Operational and clinical research		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77004	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	UNSOL-JSC
Start Date:	01/01/2006	**End Date:**	12/31/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Pellis, Neal	**Contact Phone:**	
Contact Email:	neal.r.pellis@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 12/31/2007 per PI (12/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNJ06HA40G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The utility of the above project is apparent, in view of the high level of importance placed by NASA on understanding and developing countermeasures to the effects of microgravity exposure and consequent unloading-induced bone loss and immune suppression in astronauts. Both these projects could provide novel information about the effects of unloading on the complex cellular interactions between cells of the skeletal system and the potential use of pharmacological countermeasures such as statins and bisphosphonates. The signaling experiments would delineate target molecules which could be manipulated to maintain equilibrium even in unloaded conditions. Cellular and tissue level changes occur much earlier than whole organ changes, which can be addressed in both these projects, thereby providing a mechanistic understanding (cell signaling modulation within the bone matrix/immune cell complex) and basis for developing whole body countermeasures for alleviating bone loss and immune suppression from the space flight		

perspective. Both these projects will involve travel to national and international NASA related and basic science related symposia by the PI.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

The project was slated to begin in January and is funded from Jan-Dec 2006. It is in its fourth month. The PI began to work on protocol optimization and experimental set up. The protocols for immunofluorescence and cell culture have been completed. The request for the antibody from University of Southwestern Wales has been sent with appropriate letters to the US dept of customs, regarding antibody storage. One graduate student has been recruited for the project with recruitment of two summer interns and one other graduate student from Texas Southern University. The summer interns will start at the end of May 2006.
Two sets of experiments with normal human lymphocytes were completed in March 2006. Total RNA is being isolated from these samples. Primers for key upstream lymphocyte targets were prepared by the PI after extensive gene bank data base searches and primer preparation strategies. The primers were then prepared using a DNA synthesizer. After completion of some protein experiments which were promising and reproducible, an abstract was submitted to the "Society of "in vitro" Biology" and accepted for a presentation in June, 2006.

This summer will entail detailed training of the students in laboratory techniques and begin proposed experiments. There have been two major hurdles in the project. The PI has been unsuccessful in acquiring regular access to JSC facilities where the project is to be based and has had to complete some experiments at TSU and the UT Medical School. Work is ongoing to rectify this but there is no news at the present moment. The other major issue has been revamping of facilities at JSC and lack of incubators and hoods for tissue culture until March 2006.

Citation: Signal transduction targets of modeled microgravity:

ALAMELU SUNDARESAN1 (Lalita) , Kamleshwar Singh1 , Neal.R.Pellis2 and 1James DuMond Jr .1Texas Southern university, Houston, TX, , 2 Human Research Program, NASA Johnson Space Center, Houston TX,: E-mail;sundaresana@TSU.edu

Abstract:

Lymphocyte signal transduction is of paramount importance for an adequate mounting of the immune response. Defects in this area could leave the host prone to infection etc. From activation to the final function be it cell proliferation, locomotion or apoptosis, the signal transduction cascade is critical. Previous experiments demonstrated that both lymphocyte activation and locomotion were inhibited in spaceflight and in ground based modeled microgravity (MMG) cultured human lymphocytes. Calcium signaling pathways were functional in MMG. In T cell activation at the membrane level, Phospholipase C gamma1 (PLC-gamma1) facilitates hydrolysis of phosphoinositide di phosphate (PIP2) to DAG and a molecule called IP3 (inositide tri phosphate). DAG activates the PKC signaling cascade and IP3 drives the calcium pathways. While examining human lymphocytes for status of different signaling molecules by immunoprecipitation and, western blotting, deficiencies were found in activation of PLC-gamma1, this being the predominant isoform active in T cells, is activated by tyrosine phosphorylation. The activated form of the enzyme was found to be down-regulated by more than two fold in MMG cultured lymphocytes. Calcium independent isoforms of PKC such as PKC delta and epsilon were also downregulated by more than 50% in MMG compared to ground cultured lymphocytes. This was observed both at the transcriptional and translational levels by immunoblotting and RT-PCR. The IP3 receptor, whose expression is indicative of calcium mobilization, however did not change in MMG cultured cells. This corroborates evidence using ionomycin that calcium signaling might be unaltered in MMG. MMG thus affects certain signaling cascades selectively. Mouse lymphocytes cultured in modeled microgravity also demonstrated decreased activation compared to ground controls. IP3 receptor expression in MMGcultured mouse lymphocytes also was not significantly different as compared to ground cultured lymphocytes. IP3 receptor levels were found to increase in expression as seen by immunobloting in MMG cultures. Nucleotide augmentation in MMG cultured mouse lymphocytes both in vitro (culture media) as well as in vivo as nutrition supplements in antiorthostaically (AOS) suspended mice, was able to recover activation both in lymphocytes isolated from control mice and from. Experiments outlining signal transduction in human vs mouse lymphocytes in MMG would go a long way in delineating immune suppression related phenomena both in space an here on earth. Nucleotide supplementation experiments would be instrumental in design of countermeasures against immunological problems incurred in space. This is especially relevant when long-term space travel is being contemplated. Results from signal transduction experiments in human and mouse lymphocytes as wells as nucleotide supplementation experiments are presented in this study.

Tracking Code: V1014-Society for In Vitro Biology, selected for presentation at the 2006 SIVB congress-June 2006, Minnesota.

Bibliography Type: Description: (Last Updated: 07/02/2026)

Abstracts for Journals and Proceedings

SUNDARESAN A (Lalita) , Singh K , Pellis NR, DuMond Jr J. "Signal transduction targets of modeled microgravity." Society for In Vitro Biology, selected for presentation at the 2006 SIVB congress-June 2006, Minnesota.

Society for In Vitro Biology, to be published in June 2006 , Jun-2006